

In the Risk Analysis matrix, the firewall is considered a high-risk network device, which indicates that you should monitor it in real time. From the Approving Security Changes section, you see that you should monitor for any changes to the firewall. This means that the SNMP polling agent should monitor such things as failed login attempts, unusual traffic, changes to the firewall, access granted to the firewall, and connections setup through the firewall.

Following this example, create a monitoring policy for each area identified in your risk analysis. It's recommended to monitor low-risk equipment weekly, medium-risk equipment daily, and high-risk equipment hourly. If you require more rapid detection, monitor on a shorter time frame. Last, your security policy should address how to notify the security team of security violations. Often, your network monitoring software will be the first to detect the violation. It should trigger a notification to the operations centre, which in turn should notify the security team, using a pager if necessary.

3. Response

Response can be broken into three parts: security violations, restoration, and review.

Security Violations: When a violation is detected, the ability to protect network equipment, determine the extent of the intrusion, and recover normal operations depends on quick decisions. Having these decisions made ahead of time makes responding to an intrusion much more manageable.

The first action following the detection of an intrusion is the notification of the security team. Without a procedure in place, there will be considerable delay in getting the correct people to apply the correct response. Define a procedure in your security policy that is available 24 hours a day, 7 days a week. Next you should define the level of authority given to the security team to make changes, and in what order the changes should be made.